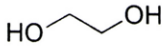
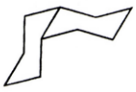
Organic Cume 03-03-2017

1. Draw the three staggered conformations of ethylene glycol shown below. The anti conformation of ethylene glycol is not the lowest energy conformation. The other two conformations are actually lower in energy than the anti conformation. Suggest an explanation.



ethylene glycol

2. Consider the structures of cis-decalin and trans-decalin:

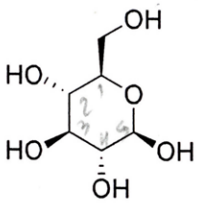


cis-decalin

trans-decalin

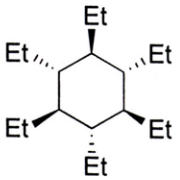
(a) Which of these compounds would you expect to be more stable? Explain your choice.
(b) One of the two compounds is incapable of ring flipping. Identify it and explain your choice.

3. Glucose (a sugar) is produced by photosynthesis and used by cells to store energy. Draw the most stable conformation of glucose:

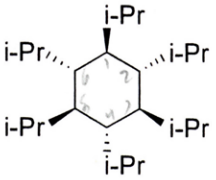


Glucose

4. The all-trans-1,2,3,4,5,6-hexaethylcyclohexane (**1**) prefers the all-equatorial conformation while the all-trans-1,2,3,4,5,6-hexaisopropylcyclohexane (**2**) possesses a severely destabilized all-equatorial conformation.

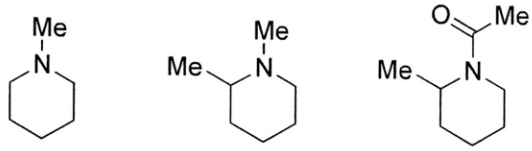


1



2

(a) Explain why adjacent equatorially oriented isopropyl groups experience severe steric interactions that are lacking in the ethyl case. Draw a chair conformation of the former case that illustrates these severe steric interactions. Also draw a Newman projection looking down one of the C–C bonds connecting the cyclohexyl ring to an equatorial isopropyl group and illustrate a conformation with severe steric strain.
(b) The all-axial conformation of **2** possesses a fairly stable conformation where the steric repulsion among the axial isopropyl groups can be minimized. Draw this

chair conformation. Also, draw a Newman projection looking down one of the C–C bonds connecting the cyclohexyl ring to an axial isopropyl group and illustrate why this conformation is lower than the all-equatorial chair conformation in steric destabilization.

5. Draw the lowest energy conformation of the following amines/amides.